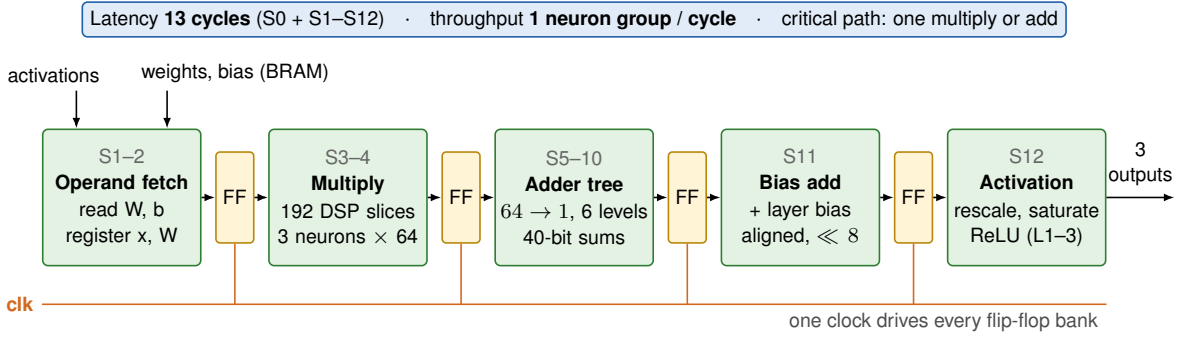
Latency 13 cycles (S0 + S1–S12) · throughput 1 neuron group / cycle · critical path: one multiply or add
activations
weights, bias (BRAM)
S1–2
Operand fetch
read W, b
register x, W
FF
S3–4
Multiply
192 DSP slices
3 neurons × 64
FF
S5–10
Adder tree
64 → 1, 6 levels
40-bit sums
FF
S11
Bias add
+ layer bias
aligned, ≪ 8
FF
S12
Activation
rescale, saturate
ReLU (L1–3)
3
outputs
clk
one clock drives every flip-flop bank